# 6.1.1- 6.1.2

Sunday, April 26, 2026 8:22 AM

*Exercise* 96. Define a decision function such that the associated language $L_{Exercise_1}$ consists of all solutions to the equation $5x+4=28+2x$ over $\mathbb{F}_{13}$. Provide a constructive proof for the claim: "There exists a word in $L_{Exercise_1}$, and verify the proof.

*Exercise* 97. Consider modular 6 arithmetic ($\mathbb{Z}_6$) from example 11, the alphabet $\Sigma = \mathbb{Z}_6$ and the following decision function:

$$R_{example_11} : \Sigma^* \to \{true, false\} \; ; < x_1, \ldots, x_n > \mapsto \begin{cases} true & n = 1 \text{ and } 3 \cdot x_1 + 3 = 0 \\ false & else \end{cases}$$

Compute all words in the associated language $L_{example_11}$, provide a constructive proof for the statement "There exist a word in $L_{example_11}$" and verify the proof.

96 alphabet $\Sigma = \mathbb{F}_{13}$

$R_{exercise_1} : (\mathbb{F}_{13})^* \to \{true, false\} ; < x_1, \ldots, x_n > \mapsto \begin{cases} true & n=1 \text{ and } 5x+4 = 28+2x \\ false & else \end{cases}$

$x = 8$

$R_{exercise_1}(<8>) = true$

97. $3x + 3 = 0$ over $\mathbb{Z}_6$

$3x = 3$

| · | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 2 | 3 | 4 | 5 |
| 2 | 0 | 2 | 4 | 0 | 2 | 4 |
| 3 | 0 | 3 | 0 | 3 | 0 | 3 |
| 4 | 0 | 4 | 2 | 0 | 4 | 2 |
| 5 | 0 | 5 | 4 | 3 | 2 | 1 |

$x \in \{1, 3, 5\}$

$R_{example_11}(<1>) = true$

*Exercise* 98. Consider the modular 6 arithmetic $\mathbb{Z}_6$ from example 11 as the alphabets $\Sigma_I$ and $\Sigma_W$, and the following decision function:

$$R_{linear} : \Sigma^* \times \Sigma^* \to \{true, false\} \; ;$$
$$(i; w) \mapsto \begin{cases} true & |i| = 3 \text{ and } |w| = 1 \text{ and } i_1 \cdot w_1 + i_2 = i_3 \\ false & else \end{cases}$$

Which of the following instances $(i_1, i_2, i_3)$ has a proof of knowledge in $L_{linear}$?

$$(3,3,0), \quad (2,1,0), \quad (4,4,2)$$

*Exercise* 99 (Edwards Addition on Tiny-jubjub). Consider the Tiny-jubjub curve together with its twisted Edwards addition law from example 71. Define an instance alphabet $\Sigma_I$, a witness alphabet $\Sigma_W$, and a decision function $R_{add}$ with associated language $L_{add}$ such that a string $(i; w) \in \Sigma_I^* \times \Sigma_W^*$ is a word in $L_{add}$ if and only if $i$ is a pair of curve points on the Tiny-jubjub curve in Edwards form, and $w$ is the sum of those curve points.

Choose some instance $i \in \Sigma_I^*$, provide a constructive proof for the statement "There is a witness $w \in \Sigma_W^*$ such that $(i; w)$ is a word in $L_{add}$", and verify that proof. Then find some instance $i \in \Sigma_I^*$ such that $i$ has no knowledge proof in $L_{add}$.

98 $i_1 \cdot w_1 + i_2 = i_3$ over $\mathbb{Z}_6$

$3w_1 + 3 = 0$ | $2w_1 + 1 = 0$

$3w_1 = 3$ | $2w_1 = 5$

$3w_1 + 3 = 0$
$3w_1 = 3$

| $\cdot$ | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 2 | 3 | 4 | 5 |
| 2 | 0 | 2 | 4 | 0 | 2 | 4 |
| 3 | 0 | 3 | 0 | 3 | 0 | 3 |
| 4 | 0 | 4 | 2 | 0 | 4 | 2 |
| 5 | 0 | 5 | 4 | 3 | 2 | 1 |

$x \in \{1, 3, 5\}$

$(3, 3, 0)$ ✓

$2w_1 + 1 = 0$
$2w_1 = 5$

$\emptyset$

$(2, 1, 0)$ ×

$4w_1 + 4 = 2$
$4w_1 = 4$

$x \in \{1, 4\}$

$(4, 4, 2)$ ×

99 *Example* 71 (The Tiny-jubjub curve). Consider the prime field $\mathbb{F}_{13}$ from exercise 50. If we choose $a = 8$ and $b = 8$, then $4a^3 + 27b^2 \equiv 6 \pmod{13}$, and the corresponding elliptic curve is given by all pairs $(x, y)$ from $\mathbb{F}_{13}$ such that $y^2 = x^3 + 8x + 8$ holds. We call this curve the **Tiny-jubjub** curve (in its affine Short Weierstrass representation), or *TJJ_13* for short.

Since the set $\mathbb{F}_{13} \times \mathbb{F}_{13}$ of all pairs $(x, y)$ from $\mathbb{F}_{13}$ only contains $13 \cdot 13 = 169$ pairs, we can compute the curve by just inserting every possible pair $(x, y)$ into the Short Weierstrass equation $y^2 = x^3 + 8x + 8$. We get the following result:

$$\begin{aligned} TJJ_13 = \{&\mathcal{O}, (1,2), (1,11), (4,0), (5,2), (5,11), (6,5), (6,8), (7,2), (7,11), \\ &(8,5), (8,8), (9,4), (9,9), (10,3), (10,10), (11,6), (11,7), (12,5), (12,8)\} \end{aligned} \tag{5.2}$$

Twisted edward curve

$$E(\mathbb{F}) = \{(x, y) \in \mathbb{F} \times \mathbb{F} \mid a \cdot x^2 + y^2 = 1 + d \cdot x^2 y^2\} \tag{5.33}$$

$$TJJ_13_E = \{(x, y) \in \mathbb{F}_{13} \times \mathbb{F}_{13} \mid 3 \cdot x^2 + y^2 = 1 + 8 \cdot x^2 \cdot y^2\} \tag{5.36}$$

Now that we have the abstract definition of our Tiny-jubjub curve in Twisted Edwards form, we can compute the set of points by inserting all pairs $(x, y) \in \mathbb{F}_{13} \times \mathbb{F}_{13}$, similarly to how we computed the curve points in its Short Weierstrass or Edwards representation. We get the following:

$$\begin{aligned} TE_TJJ_13 = \{&(0,1), (0,12), (1,2), (1,11), (2,6), (2,7), (3,0), (5,5), (5,8), (6,4), \\ &(6,9), (7,4), (7,9), (8,5), (8,8), (10,0), (11,6), (11,7), (12,2), (12,11)\} \end{aligned} \tag{5.37}$$

alphabet $\Sigma_i = F_{13}$ $\quad i = (x_1, y_1, x_2, y_2) \in \mathbb{Z}_i^4$

$\Sigma_w = F_{13}$ $\quad w = (x_3, y_3) \in \mathbb{Z}_w^2$

To see what the Twisted Edwards group law looks like, let $(x_1, y_1)$, $(x_2, y_2)$ be two points on an Edwards curve $E(\mathbb{F})$. The sum of those points is then given by the following equation:

$$(x_1, y_1) \oplus (x_2, y_2) = \left( \frac{x_1 y_2 + y_1 x_2}{1 + d x_1 x_2 y_1 y_2}, \frac{y_1 y_2 - a x_1 x_2}{1 - d x_1 x_2 y_1 y_2} \right) \tag{5.38}$$

$R_{add} : \Sigma_i^* \times \Sigma_w^* \to \{true, false\}$ ;

$(i, w) \longmapsto \begin{cases} true & |i| = 4 \text{ and } |w| = 2 \text{ and } (i_1, i_2), (i_3, i_4) \in TJJ_13_E \text{ and } w = (i_1, i_2) \oplus (i_3, i_4) \\ false & \text{else} \end{cases}$

$L_{add} = \{(i, w) \in \Sigma_i^* \times \Sigma_w^* \mid R_{add}(j; w) = true\}$

Constructive proof

$$
\begin{aligned}
(5,5) \oplus (8,5) &= \left( \frac{5\cdot 5 + 5\cdot 8}{1 + 8\cdot 5\cdot 8\cdot 5\cdot 5}, \frac{5\cdot 5 - 3\cdot 5\cdot 8}{1 - 8\cdot 5\cdot 8\cdot 5\cdot 5} \right) \\
&= \left( \frac{12+1}{1+5}, \frac{12-3}{1-5} \right) \\
&= \left( \frac{0}{6}, \frac{12+10}{1+8} \right) \\
&= \left( 0, \frac{9}{9} \right) \\
&= (0,1)
\end{aligned}
$$

no instance $i \in \mathbb{Z}_1^*$ has no knowledge

proof in Ladd